

Qiran Jia

Los Angeles | qiranjia@usc.edu | github.com/BruceResearch | [LinkedIn](#) | 8056184338

EDUCATION

Ph.D. in Biostatistics - University of Southern California *August 2022- December 2026 (expected)*
M.Sc. in Biostatistics - New York University *September 2020-May 2022*
B.A. in Statistics - University of California, Santa Barbara *September 2015-December 2019*

EXPERTISE

Statistical inference; Bayesian modeling; integrative multi-omics data analysis; single-cell transcriptomics analysis; spatial omics analysis; medical image analysis; machine learning and statistical methods for high-dimensional data; deep learning approaches for neuroimaging

SKILLS

Languages: Proficient in Python (libraries: NumPy, JAX, Matplotlib, PyTorch, Scanpy), R (Seurat)

Tools: High-Performance Computing Cluster, Linux, GPU(CUDA), Git

EXPERIENCE

USC Department of Population and Public Health Sciences *September 2022-now*

- Developed a **machine learning & statistical** model, LUCID, for **integrative clustering**, association inference, and risk prediction in multi-omics data, incorporating environmental exposomes and phenotypic information. Enhanced the R package's speed by **120x** and extended functionalities, enabling different **multi-omics integration strategies**, regularization, **missing data imputation**, and visualizations.
- Proposed **Praxis-BGM**, a **semi-supervised transfer-learning-based** clustering approach using **variational inference** for omics data, incorporating multiple sources of prior information, including pathways, to improve efficiency, estimation accuracy, and interpretability of the clusters. The method has been extended to **single-cell and spatial transcriptomics** (e.g., Xenium), enabling effective **label transfer** across datasets.
- Provided statistical consulting for environmental health and epidemiology labs on various projects investigating associations between **environmental exposures** (e.g. PFAS) and health outcomes mediated by multi-omics data.
- Collaborated with interdisciplinary researchers at AIMS@USC to develop computational methods and pipelines for **single-cell spatial omics analysis**, including **cell segmentation**, **cross-dataset label transfer**, and **multimodal data integration**, to move beyond rule-based approaches.

NYU School of Global Public Health *February 2021-September 2022*

- Supported end-to-end **model development, evaluation, and data analysis** for **neuroimaging-focused** research projects.
- Designed and implemented a **deep learning model** combining **convolutional neural networks and vision transformers** for **3D medical image segmentation**, improving model capacity to capture both local and global features.
- Developed a scalable **voxel-level multiple testing** framework using fully connected hidden Markov random fields, enabling **full-brain FDG-PET analysis** with **hundreds of thousands of voxels** and improving power in detecting Alzheimer's disease regions.

PUBLICATIONS

Bitr-UNet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation

Q. Jia, H. Shu

International MICCAI Brainlesion Workshop, pp. 3-14, 2021

MFI-PSO: A Flexible and Effective Method in Adversarial Image Generation for Deep Neural Networks

H. Shu, R. Shi, **Q. Jia**, H. Zhu, Z. Chen

2022 International Joint Conference on Neural Networks (IJCNN), pp. 01-08, 2022

Integrating Multi-Omics with Environmental Data for Precision Health: A Novel Analytic Framework and Case Study on Prenatal Mercury-Induced Childhood Fatty Liver Disease

J.A. Goodrich, H. Wang, **Q. Jia**, N. Stratakis, Y. Zhao, L. Maitre, M. Bustamante, et al.

Environment International, 108930, 2024

An Extension of Latent Unknown Clustering Integrating Multi-Omics Data (LUCID) Incorporating Incomplete Omics Data

Y. Zhao, **Q. Jia**, J. Goodrich, B. Darst, D.V. Conti

Bioinformatics Advances, 2024

DeepFDR: A Deep Learning-based False Discovery Rate Control Method for Neuroimaging Data

T. Kim, H. Shu, **Q. Jia**, M.J. de Leon

Proceedings of Machine Learning Research, 238, 946, 2024

The Potential Mediating Role of the Gut Microbiome and Metabolites in the Association Between PFAS and Kidney Function in Young Adults: A Proof-of-Concept Study

H.E. Hampson, S. Li, D.I. Walker, H. Wang, **Q. Jia**, et al.

Science of The Total Environment, 954, 176519, 2024

A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data

T. Kim, **Q. Jia**, H. Shu, M.J. de Leon

arXiv preprint, arXiv:2505.20688, 2025

Integrated Spheroid-to-Population Framework for Evaluating PFHpA-Associated Metabolic Dysfunction and Steatotic Liver Disease

B.O. Baumert, A.C. Maretti-Mira, D.I. Walker, H. Wang, **Q. Jia**, et al.

Research Square, rs.3.rs-5960979, 2024

LUCIDus: An R Package for Implementing Latent Unknown Clustering by Integrating Multi-omics Data (LUCID) With Phenotypic Traits

Y. Zhao, **Q. Jia**, J.A. Goodrich, D.V. Conti

R Journal, 2024

PRESENTATIONS

BiTr-Unet: a CNN-Transformer Combined Network for MRI Brain Tumor Segmentation for BrsTS2021

Brian Lesion (BrainLes) Workshop 2021, Virtual, September 2021

Praxis-BGM: A Prior-Augmented and Regularized Natural-Gradient Variational Inference on Gaussian Mixture Models

Joint Statistical Meetings (JSM) 2025, Nashville, TN, August 2025